

50 Networking Concepts Every Developer Should Know

The concepts behind how applications communicate

Networking Fundamentals

1. DNS

Translates domain names into IP addresses.

2. IP Address

Identifies a device or network interface on a network.

3. Port

Identifies a specific service or application on a device.

4. Packet

A small unit of data sent across a network.

5. Latency

The time it takes data to travel between two points.

Network Protocols

6. TCP

Provides reliable, ordered delivery of data.

7. UDP

Sends data without guaranteeing delivery or ordering.

8. HTTP

A protocol for transferring data between clients and servers.

9. HTTPS

HTTP secured using TLS encryption.

10. TLS

Secures communication through encryption and helps authenticate the server.

How Networks Connect

11. Router

Forwards packets between different networks.

12. Switch

Connects devices within the same network.

13. Gateway

Provides a path between different networks.

14. NAT

Translates between private and public IP addresses.

15. DHCP

Automatically assigns network configuration to devices.

Web Infrastructure

16. DNS Resolver

Finds the IP address associated with a domain.

17. CDN

Delivers content from servers closer to users.

18. Load Balancer

Distributes traffic across multiple servers.

19. Reverse Proxy

Receives requests and forwards them to backend servers.

20. Proxy Server

Acts as an intermediary between a client and another server.

Application Communication

21. API

An interface that allows software systems to communicate.

22. REST

An architectural style commonly used for web APIs.

23. WebSocket

Enables persistent, two-way communication.

24. Webhook

Sends data when a specific event occurs.

25. CORS

Controls cross-origin requests made by web browsers.

Network Performance

26. Bandwidth

The maximum amount of data a connection can transmit.

27. Throughput

The actual amount of data successfully transferred over time.

28. Timeout

Stops waiting when a network operation takes too long.

29. Retry & Backoff

Retries failed requests while increasing the wait between attempts.

30. Connection Pooling

Reuses existing connections instead of creating new ones.

Network Architecture

31. Subnet

Divides a network into smaller logical networks.

32. IPv4

A 32-bit IP addressing system.

33. IPv6

A 128-bit IP addressing system with a much larger address space.

34. MAC Address

A hardware-level identifier for a network interface.

35. Socket

A communication endpoint used by applications to exchange data over a network.

Network Security

36. Firewall

Controls network traffic using security rules.

37. VPN

Creates an encrypted connection over a network.

38. ACL (Access Control List)

Defines which traffic or users are allowed or denied access.

39. WAF (Web Application Firewall)

Protects web applications by filtering malicious HTTP traffic.

40. DDoS

Overwhelms a service with excessive traffic to make it unavailable.

HTTP Concepts

41. HTTP Headers

Carry additional information about requests and responses.

42. Cookies

Browser-stored data sent with matching requests.

43. HTTP Methods

Define what action a client wants to perform.

44. HTTP Status Codes

Tell the client how a request was handled.

45. Keep-Alive

Allows connections to be reused for multiple requests.

Network Operations

46. ARP

Maps an IP address to a MAC address on a local network.

47. TCP Handshake

Establishes a TCP connection between two endpoints.

48. TLS Handshake

Establishes a secure connection and negotiates encryption.

49. Network Monitoring

Tracks network health, performance, and failures.

50. Observability

Uses logs, metrics, and traces to understand system behavior.

Every application depends on the network.

Understanding what happens between the client and the server makes you a better developer.

 **Save this guide for the next time you debug a network issue.**

 **Comment “NETWORK” if this helped.**